Adaptive Nonlinear MPC for Efficient Trajectory Tracking Applied to Autonomous Mining Skid-Steer Mobile Robots

Alvaro Javier Prado
Departamento de Automatización y Control Industrial
Escuela Politécnica Nacional (EPN)
Quito, Ecuador
alvaro.prado.5@sansano.usm.cl

Danilo Chávez, Oscar Camacho
Departamento de Automatización y Control Industrial
Escuela Politécnica Nacional (EPN)
Quito, Ecuador
danilo.chavez;oscar.camacho@epn.edu.ec

Miguel Torres-Torriti
Departamento de Ingeniería Eléctrica
Pontificia Universidad Católica de Chile (PUC)
Santiago, Chile
mtorrest@ing.puc.cl

Fernando Auat Cheein
Departamento de Ingeniería Electrónica
Universidad Técnica Federico Santa Maria (UTFSM)
Valparaíso, Chile
fernando.auat@usm.cl

***Abstract*—The heterogeneous nature of the navigation surface suggests adaptation capabilities in vehicle motion control to overcome the effects of the wheel-terrain interaction. In such scenario, this paper presents an integral adaptive control framework built upon a Nonlinear Moving Horizon Estimator and a Nonlinear Model Predictive Control scheme, under which the objective is to on-line estimate states and model parameters of a robot motion model while autonomously navigating in off-road terrain conditions. With an adjustable model, the controller is made adaptive against terrain changes while tracking prescribed trajectories. The system is composed by two coupled subsystems to represent the vehicle motion and tire slip dynamics. The combined control-estimation strategy works under the Real-Time Iteration scheme to attain reliable computational activity for high-speed tire dynamics (e.g., slip). Trials in a simulation and real test environment with a compact mini-loader Cat® 262C, as those found in the mining industry, showed that the approach is able to efficiently estimate states and model parameters without exceeding constraints. The analysis of computational efficiency in various hardware configurations is also provided, exhibiting that the rapid optimization involved in the proposed controller is possible for high-speed dynamics.**

***Keywords*—Nonlinear Moving Horizon Estimation, Nonlinear Model Predictive Control, Real-Time Iteration, Wheel-Terrain Interaction**

Pursuing the call for the mining industry to cope with future material production and meet global requirements, new developments are pushing this activity forward the optimization and adaptation of machines in the mining field to improve productivity and provide cost-effective alternatives to skilled labour [1]. In this sense, autonomous machines arise as a potential solution for efficient transport systems that involve exploration, abstraction and transfer of material among operating spots; however, the variability of the terrain usually degrades mobility performance. In such case, adaptive control approaches for autonomous navigation under varying terrain conditions are more than welcome.

It is widely beneficial for adaptive control strategies (essentially to those based on models) to fully recover the system knowledge, and then readjust the model to new current vehicle dynamic conditions [2]. For instance, feedback control schemes, such as those employed in Model Predictive Control (MPC), are usually designed under the assumption of certainty-equivalent models; however, the unavoidable presence of uncertainties may lead to erroneous predictions and limited performance. Hence, the model adaptability through state and model parameter estimation can provide complementary robust control performance.

In practical applications, not all system states are available to feedback or some model parameters are accessible, introducing further challenges if the model stands for strongly coupled nonlinear dynamics. In fact, to avoid additional computations due to the model complexity, several schemes typically consider that all states and parameters are initially known or remain constant over time. Several estimation methods have been proposed in the literature to deal with such issue [3, and its references]. For instance, estimators for nonlinear models, under the assumption of normally distributed states and measurement noise, have been derived from multiple variants of the Extended Kalman Filter (EKF) [4]. Such approaches rely only on an unconstrained, sampled nonlinear model at a selected operation point –or around a operation set as in the case of the Unscented Kalman Filter (UKF) [5].

Unlike the aforementioned methods, Particle Filtering (PF) does not require assumptions on the probability distribution associated to an uncertain system; however, it could become impracticable for fast dynamics or unsolvable due to its intrinsic problem related to the *Course of the Dimensionality* arisen by large-sampled systems. In addition, constraints on the

978-1-7281-9365-6/20/$31.00 ©2020 IEEE

system states and model parameters can hardly be incorporated within the estimation framework.

As a suitable alternative, the Nonlinear Moving Horizon Estimation (NMHE) framework can provide real-time feasibility in the simultaneous state and model parameter estimation for fast dynamics and middle-scale problems (as those emerging from the slip phenomena). The NMHE approach uses a deterministic scheme to account for the current system evolution of a nonlinear model, and penalizes past measurement errors subject to systematic constraints on the states and model parameters. The NMHE is built upon the Real-Time Iteration (RTI) scheme, which exploits the direct multiple shooting method to access the benefits of the parallelism [6]. The potential synergy from such framework would makes of Adaptive Nonlinear Model Predictive Control (ANMPC) an appropriate combined NMHE-NMPC scheme to tackle the mobility problem uder slippery terrain conditions.

In this paper, an integral control framework is proposed under the ANMPC approach to perform trajectory tracking for autonomous navigation in changing terrain conditions. Using the RTI scheme, the combined estimation-control approach is addressed to simultaneously on-line estimate unmeasured states, adjust dynamical model parameters and adapt performance against the wheel-terrain interaction. To this end, it is raised an adjustable model of a Skid-Steer Mobile Robot (SSMR) to represent its dynamics. The main contribution of the work lies on formulating a fast estimation-control framework to overcome performance losses in off-road terrains (i.e., pavement and loose gravel).

This work is organized as follows. First, Section I formulates the SSMR motion model, including longitudinal and lateral tire slip dynamics. Section II presents the NMHE approach for state and model parameter estimation. In Section III, the architecture for the adaptive control system is elaborated and discussed. Simulation and experimental results are shown in Section IV. Finally, Section V provides the concluding remarks of the work.

I. SKID-STEER MOBILE ROBOT MODEL

The motion dynamics of a Skid-Steer Mobile Robot (SSMR) are characterized here by an adjustable dynamical model with time-varying model parameters. The model is derived from two coupled subsystems; a) a model for the vehicle motion dynamics, and b) a model for tyre slip dynamics.

A. Vehicle Motion Model

The proposed model is an extended version of an unicycle model [7]. The dive-line layout includes differential steering and a four wheel-drive traction system. The longitudinal forces are generated from the tractive force exertion and friction forces, while the lateral ones from the skidding contact point. A scheme of the SSMR model is presented in Fig. 1. The external, longitudinal, and lateral tire forces are denoted by F_e, $F_{x'}, F_{y'}$, respectively. The inertia and force balance equations with respect to the Center of Mass (CoM) are given by:

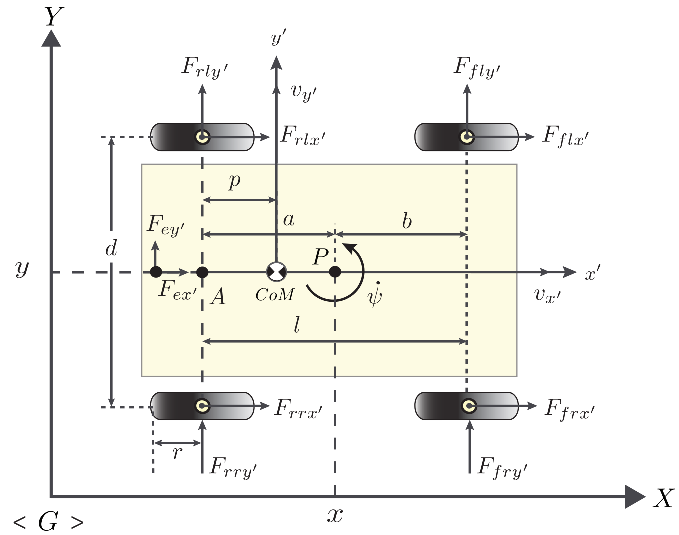

Fig. 1: Skid-Steer Mobile Robot Model. Notation depicts forces F_{ij} for each tire ij in the global reference frame G. Distances a, b, p, l and d describe the chassis arrangement.

$$
\begin{aligned}
&m(\dot{v}_{x'} - v_{y'}\omega_{z'}) = F_{frx'} + F_{rrx'} + F_{flx'} + F_{rlx'} + F_{ex'} \\
&m(\dot{v}_{y'} + v_{x'}\omega_{z'}) = F_{fry'} + F_{rry'} + F_{fly'} + F_{rly'} + F_{ey'} \\
&I_{z'}\dot{\omega}_{z'} = d(F_{frx'} + F_{rrx'} - F_{flx'} - F_{rlx'}) - p(F_{rry'} + F_{rly'}) \\
&+ (l-p)(F_{fry'} + F_{fly'})
\end{aligned} \tag{1}
$$

where the subscripts $\{f, r\}$ represent the front or rear tires, and $\{r, l\}$ for right or left tires, respectively. The robot model stands for positioning $P(x, y)$ in the $X - Y$ plane, whereas linear speeds of the chassis $v_{x'}$, $v_{y'}$, and angular speed $\omega_{z'}$ regarding the global reference frame G. The tire rotation dynamics about each wheel-motor are as follows:

$$
\begin{aligned}
\dot{v}_{ijx'} &= \frac{1}{m_w} F_{ijx} \\
J_w\dot{\omega}_{ij} + B_e\omega_{ij} &= \tau_{ij} - rF_{ijx'}
\end{aligned} \tag{2}
$$

where m_w is the quarter robot mass evenly distributed on each tire; r is the wheel radius; $F_{ijz'}$ is the vertical load distributed across the robot chassis, i.e, $F_{ijz'} = F_z$; $v_{ijx'}$ and ω_{ij} denote linear and angular speeds from each tire; τ_{ij} is the driving torque applied to each wheel axle. As the SSMR structure often normalizes torques in the actuation axles, the gearbox system only comprises traction τ_v and turning torques τ_v. Thus, the uniform torque actuation is:

$$
\tau_v = \frac{(\tau_{fr} + \tau_{fl} + \tau_{rr} + \tau_{rl})}{4}, \quad \tau_\omega = \frac{(\tau_{fr} + \tau_{fl} - \tau_{rr} - \tau_{rl})}{4} \tag{3}
$$

Each motor is often driven by hydraulic actuators, whose relationship speed-torque can be described by $\tau_{ij} = \left[v_{ijx} - v_{ijx}^{\text{ref}}\right]/k_\tau$; where k_τ is the actuator constant and v_{ijx}^{ref} is the reference speed applied to each motor actuator. For differential steering, the chassis speeds $\omega_{z'}$, $v_{x'}$ and $v_{y'}$, including the skidding speed $v^s_{y'}$, can be obtained as follows:

$$\begin{aligned}
\omega_{z'} &= \frac{1}{d}\left[\frac{r}{2}\left(\omega_{fr}+\omega_{rr}-\omega_{fl}-\omega_{rl}\right)\right] \\
v_{x'} &= \frac{1}{2}\left[\frac{r}{2}\left(\omega_{fr}+\omega_{rr}+\omega_{fl}+\omega_{rl}\right)\right] \\
v_{y'} &= \frac{p}{d}\left[\frac{r}{2}\left(\omega_{fr}+\omega_{rr}-\omega_{fl}-\omega_{rl}\right)\right]+v^s_{y'}
\end{aligned} \tag{4}$$

The motion equations of the robot chassis in the position point P with respect to G are:

$$\begin{aligned}
\dot{x} &= v_{x'}\cos\psi - [v_{y'}+(a-p)\omega_{z'}-b\omega_{z'}]\sin\psi \\
\dot{y} &= v_{x'}\sin\psi + [v_{y'}+(a-p)\omega_{z'}-b\omega_{z'}]\cos\psi \\
\dot{\psi} &= \omega_{z'}
\end{aligned} \tag{5}$$

Combining Eqs. 1 to 5, the SSMR motion model can be written as follows:

$$\begin{bmatrix}\dot{x}\\ \dot{y}\\ \dot{\psi}\\ \dot{v}_x\\ \dot{v}_y\\ \dot{\omega}_z\end{bmatrix} = \begin{bmatrix} v_x\cos\psi - a\omega_z\sin\psi \\ v_x\sin\psi + a\omega_z\cos\psi \\ \omega_z \\ \theta_0\omega_z^2+\theta_1 v_x \\ \theta_3\omega_z v_x+\theta_4\omega_z \\ \theta_6\omega_z v_x+\theta_7\omega_z\end{bmatrix} + \begin{bmatrix}0&0\\0&0\\0&0\\\theta_2&0\\0&\theta_5\\0&\theta_8\end{bmatrix}\begin{bmatrix}\tau_v\\ \tau_\omega\end{bmatrix} + [\delta_x\ \delta_y\ \delta_\psi\ \delta_{v,x}\ \delta_{v,y}\ \delta_{\omega,z}]^T \tag{6}$$

where x, y, and ψ represent the robot pose fixed at P; ω_z stands for the angular speed of the robot. The components of the robot speed v are described by the longitudinal v_x and lateral speeds v_y. The speed v_y affects the lateral motion of the robot in which nonholonomic constraints could not be met while skidding. Further, as the vector of model parameters $\theta_{sub1} = [\theta_0, ..., \theta_8]^T$ is often difficult to obtain directly, the non-dependent elements of θ_{sub1} are estimated, whereas the dependent model parameters are computed as $\theta_6 = (a-p)\theta_2$, $\theta_7 = (a-p)\theta_3$, $\theta_8 = (a-p)\theta_5$. It is worth of noting that the motion dynamics are posed by a non-linear model with additive uncertainties $\delta_{m1} = [\delta_x\ \delta_y\ \delta_\psi\ \delta_{v,x}\ \delta_{v,y}\ \delta_{\omega,z}]^T$.

B. Wheel Rotation Model

The wheel model is proposed as a function of the longitudinal $F_{ijx'}$ and lateral forces $F_{ijy'}$ using Pacejka insights [8]. The model relies on the vertical load $F_{ijz'}$ and friction coefficients μ_x and μ_y. These friction components represent the most nonlinear behaviour of the tyre, which define the slip ratios σ_{sij} and sideslip angles α_{ij}. A condensed version of the MF-tire model [8] is adapted here as follows:

$$\begin{aligned}
F_{ijx'} = \mu_x(\sigma_{sij})F_{ijz'},\ & F_{ijy'} = \mu_y(\alpha_{ij})F_{ijz'} \\
\mu_x(\sigma_{sij}) &= D_x\sin\left(C_x\arctan\left(B_x\sigma_{sij}\right)\right) \\
\mu_y(\alpha_{ij}) &= D_y\sin\left(C_y\arctan\left(B_y\alpha_{ij}\right)\right)
\end{aligned} \tag{7}$$

where $\theta_{sub2} = [B_{(\cdot)}\ C_{(\cdot)}\ D_{(\cdot)}]^T$ is the vector of tire model parameters. This model simplifies the camber influence since tires do not tilt by confident speed profiles.

The tire slip ratio σ_s and the sideslip angle α represent the wheel-terrain interaction effects since they provide an idea about the speed change of the robot by traction and skidding force losses. The slip ratio σ_s for each tire is given by:

$$\sigma_{sij} = \begin{cases} \frac{s_{ij}}{\omega_{ij}r},\ \text{if } \omega_{ij}r > v_{ijx'},\ \omega_{ij}r \neq 0,\ \text{for driving,} \\ \frac{s_{ij}}{v_{ijx'}},\ \text{if } \omega_{ij}r < v_{ijx'},\ v_{ijx'} \neq 0,\ \text{for braking,} \\ \text{with} \quad s_{ij} = \omega_{ij}r - v_{ijx'} \end{cases} \tag{8}$$

It is worth of mentioning that the tire dynamics becomes saturated as σ_s exceeds the operating range $[-1, 1]$, thus proving safety constraints on the working conditions.

The sideslip angle α represents the robot drift in the lateral direction. Since the transversal axles of tires often operate synchronously in differential turning, each pair of lateral tires result in similar skidding unless slip occurs. Herein, it is considered that: $\alpha_{fr} = \alpha_{rr}$, $\alpha_{fl} = \alpha_{rl}$, for relieving the model complexity. Then, the sideslip angles are given by:

$$\alpha_R = \alpha_{ir} = \left(\frac{v_{iry'} - b\omega_{z'}}{v_{irx'}}\right),\ \alpha_L = \alpha_{il} = \left(\frac{v_{ily'} + a\omega_z}{v_{ilx'}}\right) \tag{9}$$

After substituting Eq. 7 into Eq. 2, and replacing into derivatives of the slip speeds and sideslip angles in Eqs. 8-9, the equations of the (fast) tire slip dynamics with additive slip uncertainty $\delta_{m2} = [\delta_{s_{ij}}\ \delta_{\alpha_{ir}}\ \delta_{\alpha_{il}}]^T$ become:

$$\begin{aligned}
\dot{s}_{ij} &= -\left(\frac{r^2}{J_w}+\frac{1}{m_w}\right)D_x\sin\left(C_x\arctan\left(B_x\sigma_{sij}\right)\right)F_z \\
&\quad + \frac{r}{J_w}\tau_v + \delta_{s_{ij}} \\
\dot{\alpha}_{ir} &= -\frac{\alpha_{ir}}{m_w v_{irx'}}D_y\sin\left(C_y\arctan\left(B_y\alpha_{ir}\right)\right)F_z \\
&\quad + (c-b)\left[-\frac{1}{l}\theta_8 v_{irx'}\left(\alpha_{ir}-\alpha_{il}\right)+\frac{\theta_{13}}{v_{irx}}\tau_\omega\right]+\delta_{\alpha_{ir}} \\
\dot{\alpha}_{il} &= -\frac{\alpha_{il}}{m_w v_{ilx'}}D_y\sin\left(C_y\arctan\left(B_y\alpha_{il}\right)\right)F_z \\
&\quad + (c+a)\left[-\frac{1}{l}\theta_8 v_{ilx'}\left(\alpha_{ir}-\alpha_{il}\right)+\frac{\theta_{13}}{v_{ilx}}\tau_\omega\right]+\delta_{\alpha_{il}}.
\end{aligned} \tag{10}$$

The resulting MF-tire model is made adjustable to the terrain variability by estimating the model parameter vector $\theta = [\theta_{sub1}; \theta_{sub2}] \in \mathbb{R}^{n_\theta}$. The robot dynamics in (Eq. 6 and Eq. 10) can be compactly rewritten in the form:

$$\dot{z}(t) = f(z(t), u(t), \theta) = \begin{bmatrix} f_1(z_1, u, \theta_{sub1}) \\ f_2(z_2, u, \theta_{sub2})\end{bmatrix} + \begin{bmatrix}\delta_{m1}\\ \delta_{m2}\end{bmatrix} \tag{11}$$

$$\xi(t) = h(z(t), u(t), \theta) + \upsilon \tag{12}$$

$$z(t) = [\overbrace{x\ y\ \psi\ v_x\ v_y\ v_z\ \omega_z}^{z_1\text{: states of subsystem 1.}}\quad \overbrace{s_{ij}\ \alpha_{ir}\ \alpha_{il}}^{z_2\text{: states of subsystem 2.}}]^T \tag{13}$$

$$u(t) = [\tau_v\ \tau_\omega]^T \tag{14}$$

$$\xi(t) = [x\ y\ \psi\ v_x\ v_y\ \omega_z]^T \tag{15}$$

where t denotes continuous time; $z(t) \in \mathbb{R}^{n_z}$, $u(t) \in \mathbb{R}^{n_u}$, $\xi(t) \in \mathbb{R}^{n_\xi}$, and υ pose the state vector, control input, and output, and bounded measurement noise, respectively. The overall system is required to satisfy constraints of the form:

$$z(t) \in \mathbb{Z}(t),\ u(t) \in \mathbb{U}(t) \tag{16}$$

where $\mathbb{Z}(t) \subseteq \mathbb{R}^{n_z}$ and $\mathbb{U}(t) \subseteq \mathbb{R}^{n_u}$ are compact sets used to design the control strategy, as detailed later.

II. STATE AND MODEL PARAMETER ESTIMATION

As not all system states are available for feedback in practice, it is required to estimate only unmeasured states and unknown model parameters. Thus, the NMHE approach is adapted here to provide observations at each new measurement instance using model predictions. The NMHE approach is able to hand estimates, and particularly to incorporate state and parameter constraints into one Least-Squares (LS) optimization problem as follows:

$$\begin{aligned}
\min_{z,\theta,u} \quad & \left\| \begin{matrix} \hat{z}(t_{k-M-1}) - z(t_{k-M-1}) \\ \hat{\theta} - \theta \end{matrix} \right\|_Q^2 + \int_{t_{k-M-1}}^{t_k} \|\xi_m - \xi(t)\|_R^2 dt \\
\text{s.t.} \quad & \dot{z}(t) = f(z(t), u(t), \theta) \\
& \xi(t) = h(z(t), u(t), \theta) \\
& z(t) \in \mathbb{Z}(t), \ \forall t \in [t_{k-M-1}, t_k] \\
& u(t) \in \mathbb{U}(t), \ \forall t \in [t_{k-M-1}, t_k] \\
& \theta \in \Theta \ \forall t \in [t_{k-M-1}, t_k]
\end{aligned} \tag{17}$$

where M denotes the fixed-size estimation range of past information moving through time. The discrepancy of the first estimates $\hat{z}$ and estimated model parameters $\hat{\theta}$ with respect to the former states z and model parameters θ are recursively penalized in the moving horizon window with a symmetric positive semi-definite matrix Q. Similarly, in the second term of the estimation objective, the deviation between the current measurements ξ_m and predicted measurements ξ is minimized by the appropriate selection of the symmetric positive semi-definite matrix R. The first term in (17) is generally referred to as the arrival cost, which may grow unbounded if matrix Q is not properly defined. Then, the matrix Q is separately computed for the estimation of states and model parameters considering that both are uncorrelated, i.e., $Q = \text{diag}(Q_{n_z} \ Q_{n_\theta})^T$. The matrix for state noise covariance Q_{n_z} is determined with EKF-based updates, and set as the sensitivity associated with the Kalman covariance matrix [9]. The model parameter pseudo-covariance Q_{n_θ} is not chosen small due to time-lag between true and estimated values, but large enough for better estimation accuracy. The computation of Q_{n_z} is as follows:

$$Q_{n_z}^{\text{update}} = H(t_k)P(t_k|t_{k-1})H(t_k)^T + M(t_k)S(t_k)M(t_k)^T, \tag{18}$$

with innovation prediction matrix $P(t_k)$ given by:

$$\begin{aligned}
P(t_k|t_{k-1}) = & F(t_{k-1})P(t_{k-1}|t_{k-1})F(t_{k-1})^T \\
& + L(t_{k-1})Q(t_{k-1})L(t_{k-1})^T
\end{aligned} \tag{19}$$

where

$$\begin{aligned}
H(t_k) &= \left.\frac{\partial h}{\partial z}\right|_{\hat{z}(t_k|t_{k-1})}, \ M(t_k) = \left.\frac{\partial h}{\partial v}\right|_{\hat{z}(t_k|t_{k-1})} \\
F(t_k) &= \left.\frac{\partial f}{\partial z}\right|_{\hat{z}(t_k),u(t_k)}, \ L(t_{k-1}) = \left.\frac{\partial f}{\partial \delta}\right|_{\hat{z}(t_{k-1}|t_{k-1}),u(t_{k-1})}
\end{aligned} \tag{20}$$

Under this aforementioned consideration, the filter is available to provide unbiased estimates with minimum variance due to the nature of the Kalman updates and predictions. Since the control system relies on state estimations, the estimates are careful not to exceed their boundaries in the presence of uncertainties through robust constraints $\mathbb{Z}(t)$, which is the most beneficial assumption of the proposed estimation approach. Meanwhile, the set of model parameter constraints Θ is defined after conducting real-time tests with upper and lower limits of the independent model parameters, whereas with equality constraints for the dependent ones. Moreover, the strategy to select M comprises a trade-off between computational efficiency and estimation accuracy. Then, as the computational complexity increases with M, it is assumed small enough to obtain reasonable estimates of the fast slip dynamics and not overload computational burden.

The computational demand to solve the optimization problem in Eq. 17 is usually conservative for real-time implementation proposes,particularly for fast-evolving dynamics. However, the Real-Time Iteration (RTI) scheme deals with such issue by recursively executing the estimation procedure within a single iteration. The optimization is possible within the RTI scheme, which is performed in a preparation and estimation phase. The first phase reduces latencies between the measurement instant and the feedback, whereas the estimation phase conduct the optimization problem to prevent redundant computations. The combination of both phases are exploited here in the control structure. The reader is encouraged to refer [6] for more details about the RTI scheme.

III. NONLINEAR MPC PROBLEM FORMULATION

In this section, an NMPC approach for trajectory tracking is formulated. The optimization problem associated with the proposed controller is given by:

$$\begin{aligned}
\min_{z(\cdot),u(\cdot)} \quad & \int_{t_k}^{t_k+t_{N-1}} [J(t,z,u)]\, dt + J_N(t_{k+N}, z_N) \\
\text{s.t.} \quad & z(t_k) = \hat{z}(t_k) \\
& \dot{z}(t) = f(z(t), u(t), \theta) \\
& z_N(t_{k+N}) \in \mathbb{Z}_N \\
& z(t) \in \mathbb{Z}(t), \quad \forall t \in [t_k, t_k + t_{N-1}] \\
& u(t) \in \mathbb{U}(t), \quad \forall t \in [t_k, t_k + t_{N-1}]
\end{aligned} \tag{21}$$

with:

$$\begin{aligned}
J(t,z,u) &= \| z^{\text{ref}}(t) - z(t) \|_{Q_c}^2 + \| u^{\text{ref}}(t) - u(t) \|_{R_c}^2 \\
J_N(t_{k+N}, z) &= \| z^{\text{ref}}(t_{k+N}) - z(t_{k+N}) \|_{P_N}^2
\end{aligned} \tag{22}$$

where t_N is the control prediction horizon; Q_c and R_c are symmetric positive definite matrices to provide tuning control performance; J is part of the stage cost function that details the control objectives; J_N is the terminal penalty of the cost function with stabilizing matrix P_N. As P_N is often difficult to calculate for nonlinear systems without increasing conservativeness, it was accounted here as a design parameter. The term $\mathbb{Z}_N$ describes a constraint set at the final stage, which is provided by bounds of the robot physical limitations. The stability analysis for coupled systems can be found in [9].

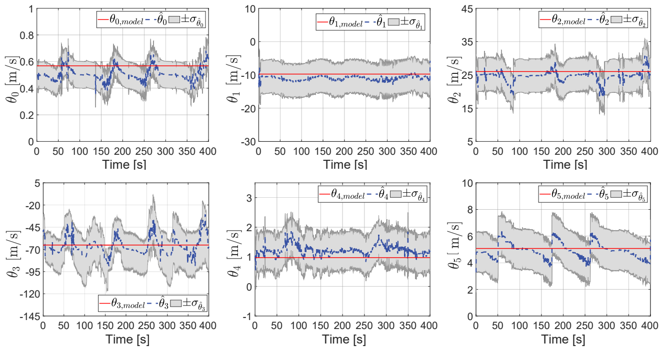

Fig. 2: Simulation results of the model parameter estimation.

TABLE I: COMPUTATIONAL PERFORMANCE IN TERMS OF REAL-TIME ITERATIONS. A FULL CYCLE UNDER THREE CPU CLOCK FREQUENCIES

		NMHE [ms]			NMPC [ms]			FCET [ms]
		EP	PP	RTI	FP	PP	RTI	
@3.2GHz	t_{min}	0.291	0.925	1.213	0.711	4.080	4.796	6.010
	t_{avg}	0.457	1.163	1.613	0.992	5.065	6.053	7.748
	t_{max}	0.824	1.933	2.758	1.311	9.125	10.453	13.217
@1.6GHz	t_{min}	1.080	2.387	3.466	2.447	15.098	17.532	21,008
	t_{avg}	1.136	2.984	4.211	5.342	17.804	23.149	28.497
	t_{max}	1.358	3.578	4.937	8.098	14.609	22.693	27.626
@1.4GHz	t_{min}	1.293	11.657	12.949	2.551	18.087	20.636	33.580
	t_{avg}	1.917	18.047	19.962	5.996	19.654	25.648	45.616
	t_{max}	2.855	20.012	22.866	9.781	17.784	27.565	47.576

* $t_{min,max,avg}$ -minimum/maximum/average computation time. EP/FP -estimation/feedback phase, PP -preparation phase. RTI -Real-Time Iteration. FCET -Full-Cycle Execution Time. Simulation parameters set to $Ts = 50ms$, $n_z = 10$, $n_\xi = 6$ $M = 15$, $N = 10$.

Notice that the global optimality is not assured here due to the problem is non-convex by nonlinear constraints.

For implementation purposes, the optimal control problem is on-line solved by using the RTI scheme (as in the estimation approach). The scheme is adapted to the receding horizon NMPC framework, where only the first element $u^*(t_k)$ of the resulting control sequence $\{u^*(t_k), ..., u^*(t_{k+N-1})\}$ is applied. For subsequent time steps, the control policy repeats the procedure using new available estimations.

IV. AUTONOMOUS NAVIGATION RESULTS

This Section presents simulation and experimental results.

A. Simulation Results

Before testing in field, the proposed ANMPC approach was assessed to verify functionality and practicality of the implementation via computational simulations. The navigation tests were conducted in gravel terrain using a designed industrial compact mini-loader model of the series Cat® 262C within the V-Rep environment. The available measurements were provided directly from the simulator, while unmeasurable states and robot model parameters were real-time estimated with the NMHE approach in parallel to the NMPC strategy, as shown in Fig. 2. By inspection, results of the parameter estimation show that the operating range of the uncertainty model is bounded, thus the model is admissible for the adjustable control strategy. Although the model parameters are not initialized with values close to the *true ones*, the estimates achieve reasonable approximations only after a few iterations within the bound set to $\sigma_\theta = \pm 15\%$.

The computational efficiency of the proposed algorithm is assessed under three different hardware configurations, i.e., Intel® Core™ CPUs @3.2GHz, @1.6GHz, and @1.4GHz. Table I shows runtimes of NMHE and NMPC, as well as the overall runtime. It is shown that for any test configuration, the FCETs do not overpass the robot sampling time set to 50ms.

Fig. 3: Skid-Steer Mobile Robot and hardware components.

B. Field Experimentations

S-shaped trajectories at constant speed profiles are pre-planned as in [7]. Two cases are reported to evaluate control performance. In the first case, the test was performed with the SSMR over pavement, whereas on loose gravel ground for the second one. The controller was implemented in a mining machine equipped with an IMU/INS VN-200 and a GPS-aided antenna for orientation angles and rotational speeds, two RTK-GPS SwiftNav Piksi® V2 for positioning and robot speeds, and two inner encoders for tire speeds, as shown in Fig. 3. More hardware details can be found in [10]. Here it was used a computer (CPU @2.2GHz) with ROS under Linux as middleware to interchange information in parallel nodes of the NMHE-NMPC framework. The controller adopts the RTI scheme from `ACADO toolkit` [6] to export C-code and solve the optimization problems of the ANMPC formulation.

Results of the cases under study are shown in Fig. 4. The field results suggest that although the SSMR is deviated from the reference trajectory due to the loose of grip in gravel terrain, the robot dynamics are rapidly compensated to overcome tracking control performance. Comparing the outcome of the first case, the second test visually shows that the robot speeds v and slip variables keep tracking faster with the ANMPC than in the NMPC approach on gravel, thus avoiding the rapid drift effect and adapting the vehicle and tire dynamics. Such compensation is clear in the tyre slip (i.e., s_L and s_R) and sideslip angles (i.e., α_L and α_R). In addition, although reasonable tracking performance can be reached in both cases under study, traction torques τ_v become more demanding to achieve adaptable capabilities in gravel terrain than in the pavement case. Then, ongoing research is aimed at fining the way to cope with such torque increase.

V. CONCLUSIONS

An integral adaptive control framework built upon Nonlinear Moving Horizon Estimation (NMHE) and Nonlinear Model Predictive Control (NMPC) has been presented and validated on a Skid-Steer Mobile Robot (SSMR) for trajectory tracking in off-road terrain surfaces. Beyond keeping real-time feasibility and practicability of the adaptive controller, the approach yielded robust performance and rapid compensation on the vehicle dynamics under pavement and disturbances of gravel terrain. Under the RTI scheme, the MHE and NMPC algorithms provided suitable optimization times considering the model complexity and fast dynamics while successfully estimating system states and model parameters. The proposed control scheme achieved effective computational performance in the millisecond range below the feedback latency.

ACKNOWLEDGMENT

This work was supported by ANID-Basal FB0008, FONDECYT 1171431, FONDECYT 1201319, FONDEQUIP 120141, CONICYT-PCHA/Doctorado Nacional/2015-21151095. Oscar Camacho thanks PII-19-03 and PIGR-19-17 Projects of Escuela Politécnica Nacional for the support regarding the realization of this work.

REFERENCES

[1] World Economic Forum, http://www3.weforum.org/docs/WEF_MM_Sustainable_World_2050_report_2015.pdf, Mining & metals in a sustainable world 2050, in: 2015 Mining and Metals in a Sustainable World, 2015.

[2] T. Kraus, E. Kayacan, J. D. Baerdemaeker, W. Saeys, High-speed adaptive nonlinear predictive control for autonomous tractor navigation, IFAC Proceedings Volumes 46 (4) (2013) 135 – 140, 5th IFAC Conference on Bio-Robotics.

[3] H. Fang, N. Tian, Y. Wang, M. Zhou, M. A. Haile, Nonlinear bayesian estimation: from kalman filtering to a broader horizon, IEEE/CAA Journal of Automatica Sinica 5 (2) (2018) 401–417.

[4] K. Iagnemma, Shinwoo Kang, H. Shibly, S. Dubowsky, Online terrain parameter estimation for wheeled mobile robots with application to planetary rovers, IEEE Transactions on Robotics 20 (5) (2004) 921–927.

[5] M. Wielitzka, M. Dagen, T. Ortmaier, Joint unscented kalman filter for state and parameter estimation in vehicle dynamics, in: 2015 IEEE Conference on Control Applications (CCA), 2015, pp. 1945–1950.

[6] B. Houska, H. Ferreau, M. Diehl, An Auto-Generated Real-Time Iteration Algorithm for Nonlinear MPC in the Microsecond Range, Automatica 47 (10) (2011) 2279–2285.

[7] J. Prado, F. Yandun, M. T. Torriti, F. A. Cheein, Overcoming the loss of performance in unmanned ground vehicles due to the terrain variability, IEEE Access 6 (2018) 17391–17406.

[8] R. Jazar, Vehicle Dynamics: Theory and Application, Springer International Publishing, 2017.

[9] E. Kayacan, W. Saeys, H. Ramon, C. Belta, J. M. Peschel, Experimental validation of linear and nonlinear mpc on an articulated unmanned ground vehicle, IEEE/ASME Transactions on Mechatronics 23 (5) (2018) 2023–2030.

[10] A. J. Prado, F. A. A. Cheein, S. Blazic, M. Torres-Torriti, Probabilistic self-tuning approaches for enhancing performance of autonomous vehicles in changing terrains, Journal of Terramechanics 78 (2018) 39 – 51.

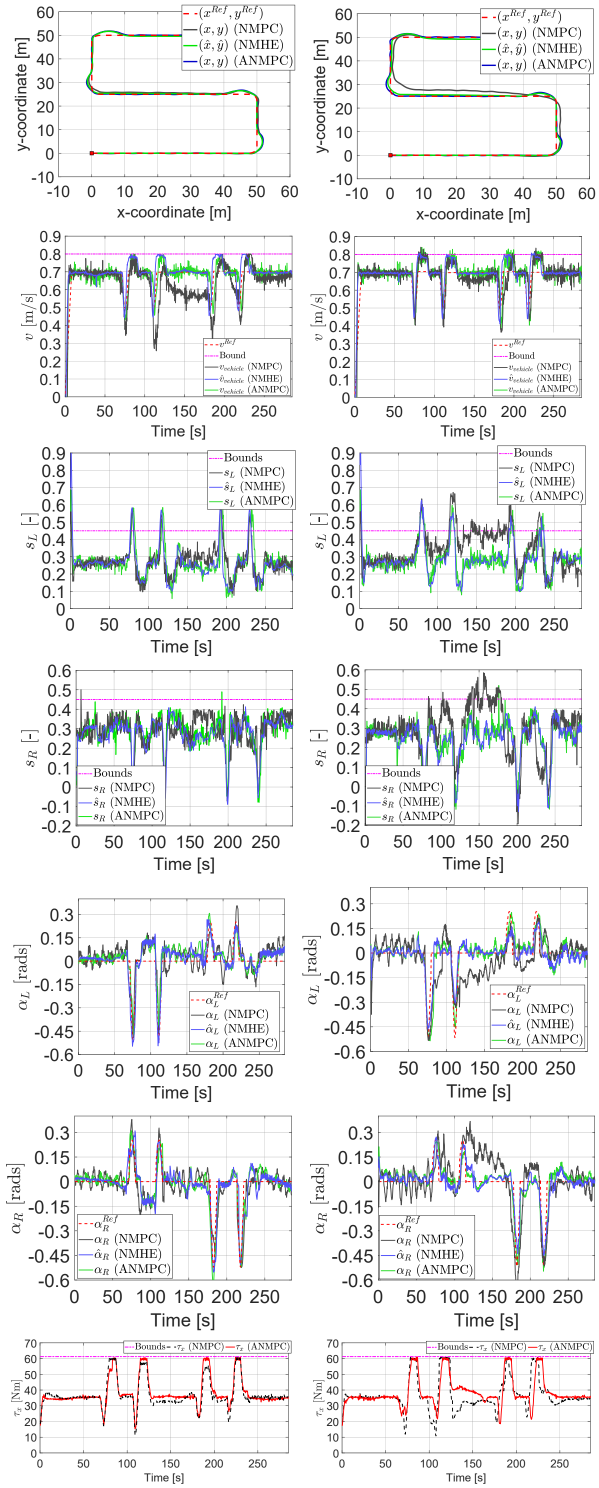

Fig. 4: Field results of the ANMPC framework while tracking S-shaped trajectories over two terrain types: pavement (left-side column) and loose gravel ground (right-side column).